

ActInf Textbook Group ~ Cohort 5 ~ Session 9

(Chapter 4, Part 2) ~ 11/14/2023

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okay well welcome back everyone it's 11:14 we're in our second discussion on chapter so before we go to the chapter and questions does anyone have any comments or thoughts otherwise we'll just try to rapidly address as many questions as possible okay onwards then so in equation 419 what is the meaning of the Tilda ETA a component of the Tilda Epsilon subv okay so tildas are trajectories Tilda ATA error minus mean minus observation or external State sometimes I remember that because the μ and ETA are like inverted of each other okay oh okay yeah yeah right but I but I I I mean there's various notation sets for internal and external States but U for mind Eda for environment exactly exactly but then e being for for error yeah but that's yeah cool okay in equation 415 are f and g associated with VF big F and EF Big G or is that just a coincidence that is just a coincidence s of Rel yeah I I mean there's there's no problem with adding more concordances but no the F and the G have nothing to do it I I think it's just f of x and then picking the next appropriate letter Jeremy if you if you want to we're just starting from the bottom just trying to hit as many of the questions as possible or feel free to bring up a a thought gotcha in the Syntax for describing [Music] expectation no I tried to make these renders a l Tech but I ended up just messing up the entire column so no it looks good I like it with the code blocks um so I can talk through this question so there's the there's the expectation and then the subscript and then the um square bracket with the function inside now no nowhere in the text could I find an explanation of the subscript and it appears to me that that subscript is more is pretty darn important like it I would I'm not clear whether that subscript contains the waiting Factor um or or what its relation is to the variables that you're summing over so like in my example I I have an X and A Y in the in the subscript P of X I think so because there are times that where that subscript contains um multiple internal variables so what I'm trying to get clear is what we are actually summing over and what the waiting factor is whether the waiting factor is included in the square brackets or not um I believe that the subscript is um what you're summing over like as if it were below the sigma

like from you know $n = 1$ to $N = 10$ and then the waiting is described by the evaluation of the function across those those categorical or continuous values right well well my my understanding of the expectation function is that you have not just you have the value at a certain in um value of x and then you have the probability of that value and so if you have a dice a six sided dice the probability is always one over six but um for our circumstances there's usually a higher weight probability right so I yeah in this block it's like I'm I'm just bothered that in that box where they explain expectation they don't explain the subscript um and it's it's causing me a lot of grief I guess okay I'll just man go for it oh did you want to add something manir okay yeah box 2.2 could have a definition of subscript um here's the probability this is the waiting here's the value of x here's the probability of X weighted by their probability and here's summing over x values and that's the expectation of X but you you you're right when you have two variables that it needs to be treated a little bit carefully okay what is meant what is meant by Niche simple simply environment of the agent there's also longer answers I'm sure people can provide anyone else want to give a thought environment of the agent partially constructed or conformed by the agent yeah maybe possible for them to to modify or construct it but but not necessarily but yes across the interaction it has to exist okay if anyone else wants to to ask a different question just raise your hand or go for it otherwise we're just trying to go to as many of the written questions as possible this is in response to the previous session asking how priors originate in addition to self-evidencing optimism bias noted in Chapter 2 I wanted to mention the notion of genes and prior genetic inheritance has been framed in active as part of Prior belief that concept itself I was was unable to find it but believe it was suggested by Paul Badcock yes um we've done evolutionary models and others have too where genes provide priors any other comments people want to make on this okay what is a generative model how is generative model related to a generative process anyone have a thought on okay so the textbook uses this distinction between a generative model and a generative process the generative Model includes the agent model of observations and also how hidden States result in those observations so internally held variables about external States um and they distinguish that from generative process which is the process that gives rise to the observations that the agent observes so that's kind of the um way that the textbook describes 2022 however very importantly in the subsequent months and years a lot of people started using generative model to refer to a system level description for example here in guest stream 45 from I guess earlier in year we had Maxwell and Ma speaking about this and in the the paper on the inner screen model of Consciousness with the quantum Fe they use generative

model now to describe the entire system um so nothing is functionally different in terms of the particular partitioning it's just about whether the generative model is referred to as the agent and then the generative model is about the generative process or the generative Model includes the whole thing and then you don't even need to mention the generative process um at 44 minutes in the transcript at this point right here Ali asks that exact question to the authors so short answer generative model is coming to be used to describe all of the system level descriptions that the model has access to and then if we're interested in just the blanket and internal States we call that the particular States or the thing if we're interested in just the internal and the active States we call those the autonomous States if we're interested in just the sensor and active states of course they're just the blanket States and then if we're interested in all of the states we can just refer to system any comments or thoughts okay I'm stuck with the derivation of equation 4 left and 413 I wrote which part I'm stuck with in the PDF cool that looks awesome um if anybody wants to go through it we could um but I think let's just save for like I'll just write a comment that it would greatly benefit from somebody who wants to look through this derivation um another nice comment would might be uh in the I believe it's in the math group one of the first links goes through um the P mdp derivation in more detail cool cool to yeah yeah okay we'll leave this one keep on going but again if anybody wants another question just raise your hand and we'll go for it otherwise just trying to give a thumbs up to as many questions as we can in equation 414 how is the necessary and sufficient condition derived why is the gradient why is the gradient of f being zero equivalent to having a policy which follows the soft Max of negative sign summation of G and F yeah this is a good question the the fully updated this is just one possible reading um when your policy posterior does not have a gradient so you're sampling your policy posterior um at a stationerity this corresponds to the soft Max um transformed negative G minus F like doing expected free energy updating of your policies is not resulting in any changes to the posterior policy distribution as scaled by the soft Max which is the shaky hands parameter So Soft Max just like it does in a neural network it sharpens or it blurs distributions so when there's a low temperature on decision making absolute zero then a policy that's slightly favored will be like upweighted a lot whereas in the high temperature decision making setting the differences amongst policies are erased but I'm not 100% sure on this one let's let's leave it okay 82 83 the equation says that we minimize free energy through action yeah we minimize free energy through two different mechanisms change the world change your mind action on the world action on the mind and only part of the free energy that depends directly on action is the

lowest level of prediction error in this setting active inference is just predictive coding plus reflex arcs what is implied here that action only happens at the reflex arc level does anyone have did anyone ask this or have a thought on this um I have a thought sure is it well in the predictive coding Paradigm um you're passing errors and so the errors are the only way that you get mind updates um so is that just just saying that um the the source of error only happens level so here in this section it's the it's a motor a motor reflex so on the on the bottom level of the hierarchy we have like the sensory motor predictive coding is doing um predictions of of sensory consequences and then reflex arcs are basically um responding just to move to just to straight without planning um or deliberation or counterfactuals the reflex arcs are just moving to resolve the difference so simple motor systems they're um able to be modeled as predictive coding on the base layer with proprioception and action plus higher level reflex arcs that basically enact default policies when the set point is deviated from how do you read interpret the higher or I key Dynamics visuals 44 and what does anyone think um one thing I would like to know is what the dot versus the arrow is insinuating that wasn't described in the L just little dot yeah yeah good question I mean one short answer is if you're interested in message passing diagrams check out the live stream 55 series because that is the most modern message passing representation and how it's being displayed in the book is just being done kind of didactically but the way that the message passing is done in live stream 55 that's like the real way if you want to study it from an engineering perspective briefly um these images show two ways of looking at hierarchical predictive processing architectures so let's just look at this one here so on the left side we have the kind of all at once hierarchy so we see a Tilda we know that Tilda means across multiple time points so what's happening here is we have these four clusters this is at level I here's level $i+1$ here's level I plus 2 so this is a three-layer predictive processing hierarchy with observations coming in from the bottom Through Time o Tilda and the highest level prior is like a fixed parameter that's how the highest hyperparameter works because if you could learn or update or Draw the highest hyper prior from another distribution then it would be another level above it so the left representation is showing the highest hyp of Prior and then the two kind of thick levels that have this four-fold Motif involving the um calculation of Errors between predicted States and incoming bottomup observations so this is the all at once view on the right that has been unfolded so now we see observation at $t-1$ t and $t+1$ so now there's three time steps from left to right and there still is the hierarchy from top to bottom any other thoughts or comments on it might want to link to the live stream or reference that as um a better source box 4.1 says the blanket of variable X

are those variables that are the uh are the variables that dot dot dot do Dot and the parents of X's children the parents of X and's children are not given a variable are these not represented the M mark on blanket formulation if they are where they show up nothing is simply given a variable whatever the whatever the node is on the base graph that's the variable used once the the system of interest is defined by the modeler then blanket States can be calculated and the Markoff blanket is everything you need to make two other sets of nodes conditionally independent so that includes not just direct parents and direct children but also co-parents I think the specific answer to the question is it shows up because the function $\mu_{\mathbf{K}}$ is taken of the function $\mathbf{K}$ uh in in the variational message par so that's saying get the parents of all the children of X μ so it doesn't need its own variable μ because those they're both functions so so getting the parents of the children is a double function call okay okay I think I see yeah yeah the parents of $\mathbf{x}$ p of X or row of X the variables are caused by X children and the parents of X's children like you could have another letter if you wanted to talk about that μ yeah short answer anytime you're actually using it you would just use the variable as defined locally what's the negative diag in equation 10 μ I think it's a Identity or a matrix diag diagonalization let's see what we wrote for line four and not not every equation has this but of course everyone is encouraged if they do want to read and help the equations to write natural language descriptions using the onology I'm not sure if this is accurate but previously we've written entropy or expected average log prob is equal to the negative diagonal Vector of dotproduct of Matrix $\mathbf{a}$ and natural log $\mathbf{a}$ is the ambiguity mapping so I think this operation helps us Define the ambig the the entropy of the ambiguity so if $\mathbf{a}$ was just the identity Matrix that would be the minimum ambiguity if because just St States observations would map cleanly to Hidden States if $\mathbf{a}$ were completely like an even Matrix like no sensory inputs could μ could be differentiated you'd have the highest entropy in equation 42 there's an equal sign with a triangle above it what does mean it means defined by definition in equation 4.8 in the second line we are breaking up the probabilities multiplying each term in expected values summing their components and so breaking into the composition of two expected values my question is why the $\tilde{Q}$ can be broken down into two $\tilde{Q}$ tildas with different arguments didn't we Define it in 4.7 as a product of $\tilde{Q}$ and $\mathbf{A}$ P why the qila broke into two CU hilders with different arguments I'm not sure which $\tilde{Q}$ Tilda is being broken up I see that $\tilde{s}$ Tilda conditioned on $\mathbf{p}_i$ and $\tilde{S}$ Tilda conditioned upon observations in $\mathbf{p}_i$ and those are the only ones that I see but the broader point about developing the μ derivations from this Top Line into all the bottom lines that sounds like good work but I'm not

sure about this question because it doesn't look like you have to split up anything in 4.10 does one sum over t to get G the right hand side has a t indicating that is calculated at particular time but G is given by the expectation over all future times does one sum over t to get G I'm not really sure which right hand side is being described here because there's so many lines the second line Okay g sub p what what about it um this may relate to the difference between T_{ao} and T So T is usually used to mean um simulation time Horizon and t to mean like agent time so here the POS the the the the G calculated on policies which could be multiple time steps long is the entropy of the Hidden States on that policy at that time plus that sequence of no it's sequence of times because you're doing a matrix operation I think it's basically the the definition of g is uh over a sequence of t time steps so yeah you're condensing the the dimension of G is can collapse because it is by definition over multiple t time steps yeah g has the same G is a functional that updates the policy prior into the policy posterior so I guess that all the time steps need to come into play in the evaluation so each of these but each of these are at a given time step but then it's I guess across time steps but this is definitely not the fullest explanation of where expected free energy is calculated equation 2.6 has a lot more information on okay I was unable to understand why the expected free energy is minimized by selecting observations that cause a large change in belief does anyone have a thought on uh there's multiple paragraphs about it but essentially look up epistemic epistemic value anywhere in the text yeah that's definitely one aspect of it which is just equation 2.6 Information Gain policies that result in sequences observations that have a large impact on inferences about hidden States Through Time ex Tilda are loaded with high Information Gain high epistemic value so expected free energy is minimized by selecting observations that cause a large change in belief yes that's half the picture expected free energy is contributed to by both the epistemic and the pragmatic value what's the role of a generative model in active inference well we talked about generative models um earlier whether we're using the kind of broader sense of generative model is describing all of the systems dependencies or the narrower sense of generative model like just the agent in relation with a generative process But whichever way you take it generative model is what we build in active inference no generative model just fun speculative imagination but this is the actual object or modeling artifact can you clarify why marginal likelihood is called marginal yes great classical history of Statistics when you have a piece of graph paper and you have um events on the rows and columns and the probabilities of those joint events in the cells on the graph paper then when you sum across the side onto the margin of the paper it's called a marginal probability okay we

have some questions with previous upvotes that we could look at or we could continue on with these ones that have no upvotes at all does anyone have any comments or thoughts or anything uh I have one comment I I'm still processing the um the the continuous derivation um section um there is one quote that has I'm trying to grock still which is I believe the mode of the let me get my text um it's talking about how here we go the motion of the mode should be the mode of the motion um in explaining the simplifications that allow the uh continuous derivation um here if you search the motion of the mode should be the mode of the motion yeah this is all related this is leading up to it okay okay um and I'm wondering if there are any sort of resources to break down um the derivation of this section in more detail similar to how there was with with the p DP par mdp yes the most relevant in basic Mechanics for stationary processes this question of hierarchical basian stationeriness arises in the generalized coordinates of motion so when you have the generalized coordinates of motion position you know and higher derivatives and you have a stationeriness in that architecture when the rate of change is zero on two different levels that are touching each other that is the situation that's being addressed here but this is I feel like if I find a graph um presentation of what's being discussed I might be able to get it so yeah I'll check that out 26 26 um here are the generalized coordinates and this is the continuous time I'm I don't remember what that yeah not sure how helpful but that at least is in the B yeah that gives me something thank you yep anyone else want to ask a question okay let's just try to do a few more basing brain helps us frame problems that an agent engaging in active inference must solve broadly these are problems of inferring states of the world perception or sense making and inferring a course of action planning or decision selection other than perception and action planning are there other tasks or challenges at the brain or organ mechanisms engage in how would we know one approach here is just to say well what are you trying to model just name your phenomena and then maybe that can be cast as a perception or as a planning task or just more broadly as a sense making or decision selection task that's possible because of course inbound and outbound statistical dependencies are the fundamental across the blanket so a huge variety of phenomena will be addressed by that but there's no reason to think that that's all that there is it just depends what kind of agent you're trying to model so it's a little bit too General to talk about all agents but are there tasks or challenges that the brain or organisms engage in just remains to be seen by somebody who wants to list out those phenomena okay for the graphical models in figure 4.2 what is an intuitive example for each structure this is a great basic question helping get familiar with the basian graph formalism like okay I boiled water and it was hot and it made a sound or I took

two ingredients and I made one dish but this is kind of the fundamentals for um all the other Bas graph discussions okay this is another great um basic or consolidating question that people could give thoughts on what are the definitions of variational free energy and expected free energy what roles do they play individually or together in active inference anyone want to give a thought otherwise this is truly a fundamental active question makes me think of like a class variational free energy equation 2.5 it's the real time gauge on how well things are going variational free energy is a functional of only prior beliefs Q variational beliefs and incoming sensory data so it computes a measure of how surprising sensory incomes are as a function of the agents beliefs at that time so it's like a diagnostic of real-time sense making expected free energy transposes that setting and that variational basian approach into counterfactuals of Futures that haven't happened yet so expected free energy considers sequences of possible observables that could happen conditioned upon different actions that the entity has the agency select amongst affordances and those policies are evaluated in terms of the epistemic and pragmatic value and then given the policy posterior the modular can take different methods for policy selection like you could just draw from the policy posterior or you could select the highest evaluated policy so everything can be understood as having a variational free energy due to its persistence through time and certain kinds of actions selection and planning agents can be understood as doing something like expected free energy and live stream 55 generalized free energy GFE it may swallow them both so we may look back on 2023 when we needed to differentiate vfe and efe it may just be a transient framing because GFE has some advantages relative to both them what is the use of categorical distributions there statistical distributions that apply to things that are categorical in the world in message passing is there a decay of information as the distance between variable X and the individual markall blanket constituents increases is implementation of information Decay and message passing an option for model implementation I remember we had some quite interesting discussions on this in a previous time any thoughts on basian graphs are topological they're not geometric so it's not exactly clear what distance means here but let's just interpret distance topologically like how many connections the information has to transfer through then you could imagine a network that's lossless at which point there would not be a decay of information or you could imagine a lossy basian graph where information was lost is impation is of information Decay implementation yes you can always develop models of information Decay however I would caution message passing is probably not where that would happen message passing um is a very lowlevel technique that probably doesn't have any advantages to Simply have

informational degradation happen in but ironically or interestingly you might want to execute a a a rigorous message passing schedule to model information Decay on a base graph however you wouldn't want to be information decay in the message passing you'd want the message passing to accurately describe the information decay of the base graph to update beliefs about policies we find the posterior that minimizes the free energy does poster at time T become a prior at time t plus one yes any thoughts or questions people want to make few more here few more down here Q is not a single variable Q is used to refer to any variational distribution of choice so it can it can take on different characteristics it can be conditioned upon different things as well you still may have to marginalize over different things or condition upon non-different things so is Q Tilda conditioned upon so is Q of X conditioned on P_i or Q of O condition on P_i a separate variational quantity which is defined in C 410 as categorical of O sub PT this one no definitely not it's a separate variational quality okay policy selection active inference automatically balances exploration exploitation already this is quite a contentious claim I mean does it automatically balance it such that it's successful or does it give us a pallet or an expressivity to set up systems that can try to balance it I I prefer the second one but it's written like the first one one the balance between epistemic and pragmatic value depends on basically among other things the intensity of C this prior is the waiting factor between exploration and exploitation yes the amplitude of C relates to how much pragmatic value is calculated for the EF and thus is one way to tune explore exploit ballots another way to do it is to modify the ambiguity of sensory States what is the difference between some other formulation for cognitive agents that places an arbitrary waiting factor between exploration and the only thing I can think of is e makes possible variational approximation methods for optimization which would be unavailable to an objective function simply awaited some of the terms yeah I mean it would be helpful to provide what cognitive architectures you're um you're thinking of for years we've we've as people have brought up other COG of architectures like I've here and elsewhere I've reviewed them um there are usually a vast number of differences between any given cognitive architectures but as this question highlights one of them would be a traditional way to approach this um problem or tension is to have an explicit waiting factor between exploration exploitation whereas in the in the ACM generative model you don't have an explicit tradeoff Factor you just have the generative model and then in different environmental settings it is going to have different arising behaviors and it can do the variational approximation methods but you could have a cognitive architecture that was like a variational autoencoder which would enable you to use variational methods so it's not that the

only difference is the variational methods any kind of last thoughts or questions I have a quick question when we say variational methods doesn't it really essentially mean that we have a some kind of uh parameterized model that we're optimizing through some kind of um optimization procedure that's so that would apply to so effectively like um deep learning um methods are in in in a sense also right they they could be and many R most narrowly it's when the approximation to exact Bas is used so not all methods use approximation so not all deep learning or statistics are are variational statistics however broad categories of are so using this variational autoencoder scheme the input obviously is the sensory put the lat and space are the cognitive States and then the output is the action selection so all of this which is otherwise like just an engineering sketch all of this is kind of refined into the first principles basian mechanics particular partition physics of cognitive systems path of least action all of those features that's the kind of analytical setup of active inference then it it but it neatly touches to existing architectures like this but again there's also machine learning architectures that don't use methods so it's not simply just being parametric this is a this is a short page but also a relevant one energy based models which is funny because they even talk about generative models so reproducing data distributions using these statistically physics based methods you know there was a there was a paper that um I don't think made uh much of a splash but it's really interesting in the context of um somewhat somewhat in the context of active inference is that uh effectively every classifier um no matter how it's trained can be seen as an energy based model and that was the title of the paper um your classifier is secretly an energy based model and and you should treat it as such um so I think there's some kind of really interesting deep connection there between um even a model that's not explicitly train using variational methods if you look at through this through the lens of that paper then obviously the equation the the equations for energy based models they look very much like um the formalism um of uh based on free energy principle uh I'm sure there's some interesting nugget of gold there to to to that that could be dug out um for someone a little bit more vers in this mathematically but it's interesting that the stuff is just hiding there yep totally agreed even when the notation and the ontology look very different there's no reason to think that that's the end of the road yeah so thank you all see you all next time thank you so much peace by